

Bank Market Power

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Motivation

- Corporate loan markets are **highly concentrated**
 - In Italy, top 5 banks in a province account for 80% of local loan market
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 - What are its **macroeconomic consequences**?

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 - What are its **macroeconomic consequences**?
- This paper: **measurement** + **GE model**
 - **Empirics**: strategy to quantify bank market power using credit register data
 - **Model**: quantify the macroeconomic implications of bank market power

The Idea in a Nutshell

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 - Large coverage, available for many countries
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Use **bankd fixed effects** to measure **deviations from law of one price**

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- We apply this core idea to the Italian credit register. Two main findings:
 1. Bank FE explain **15%–25% of the variation** in interest rates
 2. Bank FE **correlate with measures of market power** (market share, # of competitors)

Model

- Dynamic model with financial frictions (in the spirit of Gertler-Karadi-Kioytaki)
 - Firms borrow to finance working capital; Banks face **leverage constraints**
 - Key novelty: **loans are imperfectly substitutable** + **oligopolistic competition**
 - Banks charge different rates because of differences in **cost of funds** and **market shares**

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 - Untargeted: relationship btw bank FE and different measures of market power
- Counterfactual analysis
 - Costs of market power (mc pricing): **+1.2% output (SR) → -6.4% (LR)**
 - Macro-pru: hurts small banks → **raises concentration** → **endogenous rise in markups**

Literature

- Interest rate dispersion across bank-firm pairs
 - Cavalcanti, Kaboski, Martins, Santos (2023), Altavilla, Boucinha, Bouscasse (2024), Faria-e-Castro, Kozlowski, Majerovitz (2025), Ferreira, Kozeniauskas (2025), Amiti, Kashyap, Kovner, Weinstein (2026)
 - **Market power → bank FE; disciplined by GE model**
- Macroeconomic modeling of banking
 - Gertler, Kiyotaki (2010), Gertler, Karadi (2011), Corbae, d'Erasmus (2021), Ulate (2021), Bianchi, Bigio (2022), Dempsey, Faria-e-Castro (2025), Jamilov, Monacelli (2025), Villa (2025)
 - **Oligopolistic competition + rich firm-bank network to fit credit register data**
- Inefficiency costs of market power in other markets
 - Edmond, Midrigan, Xu (2023), De Loecker, Eeckhout, Unger (2020), Baqaee, Farhi (2020), Berger, Herkenhoff, Mongey (2022)
 - **Quantify the output costs of market power in credit markets**

Outline

1. Empirical Analysis
2. Model
3. Quantitative Analysis

Empirical Evidence

Empirical Analysis: Roadmap

- **Known fact:** large **dispersion in loan rates** after controlling for default risk and collateral
Cavalcanti et al. (2024), Faria-e-Castro et al. (2025), Ferreira and Kozeniauskas (2025), Amity et al. (2026)

Empirical Analysis: Roadmap

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 - Identified from **same firm paying different interest rates to different lenders**
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 - If loans are perfect substitutes → deviation from the law of one price
 - Dispersion in bank FEs may reflect service differentiation

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 - If loans are perfect substitutes → deviation from the law of one price
 - Dispersion in bank FEs may reflect service differentiation
- **Result 2:** Bank fixed effects **positively correlated with measures of market power**

Data

- **AnaCredit:** harmonized euro-area credit register (ECB) [Details](#)
 - Loan-level: interest rate, outstanding/committed amounts, loan type, maturity, collateral
 - Firm-level: bank assessment on default probability
- **Italian Credit Register:** Italian archive of interest payment information (Bank of Italy)
 - Longer time series of bank-firm relationships (2005:2019)
 - Better coverage of small local banks
- Merged with **firm balance sheets, bank balance sheets, default risk from risk-analytics firm**
- Baseline sample: overdraft accounts in 2019 (AnaCredit), Italian firms, ≥ 5 employees

Summary Statistics

	Obs.	Mean	SD	p25	p50	p75	p95	
<i>Panel A: Firm Characteristics</i>								
Employment	90,478	62.2	1,611.1	9	15	31	141	
Total assets (€M)	90,478	36.5	824.0	1.0	2.6	7.3	54.9	
Revenue (€M)	90,478	30.2	553.0	1.2	2.7	7.7	57.2	
No. bank relationships	90,478	2.3	1.6	1	2	3	6	
No. loans (all types)	90,478	4.3	18.8	1	2	4	12	
	Obs.	Mean	SD	p25	p50	p75	p95	% > 0
<i>Panel B: Short-Term Credit (€000)</i>								
Total	90,478	1,349	7,260	98	260	799	4,832	100
Liquidity credit	90,478	196	1,802	0	15	100	620	58
Irrevocable credit lines	90,478	104	839	0	0	0	400	24
Receivable-based	90,478	442	3,116	0	50	285	1,645	59
Term loans	90,478	363	3,721	0	0	25	1,200	27
Trade financing	90,478	126	1,261	0	0	0	352	10
	Obs.	Mean	SD	p25	p50	p75	p95	
<i>Panel C: Liquidity Credit (Loan Level)</i>								
Interest rate (%)	109,387	6.14	4.78	2.75	5.00	8.15	14.62	
Prob. of default	109,387	0.04	0.12	0.01	0.01	0.03	0.12	
Collateral value (%)	109,387	53.65	48.37	0	87.78	100.00	100.00	
Amt. outstanding (€000)	109,387	122	645	6	28	86	412	

AKM Decomposition

- **Step 1:** Residualize interest rates on default risk and collateral $\rightarrow \tilde{r}_{fb,t}$
 - Assessment made by bank b on default probability of firm f
 - Risk & collateral explain 26% of variation
- **Step 2:** Decompose residual $\tilde{r}_{fb,t}$ into firm and bank components

$$\tilde{r}_{fb,t} = \alpha_{f,t} + \alpha_{b,t} + \varepsilon_{fb,t}$$

- $\alpha_{b,t}$: **bank-province FE**
- Identification: firms borrowing from **multiple banks** (largest connected set)
- Key statistic: $\text{Var}(\alpha_{b,t})/\text{Var}(\tilde{r}_{fb,t})$ ($= 0$ under law of one price)

AKM Results: Variance Decomposition

- **Result:** bank FE explains a substantial share of dispersion in interest rates
 - about **half** the variation explained by default risk & collateral

No Controls	
Std. deviation	2.87
Variance Firm FE %	0.69
Variance Bank FE %	0.17
Variance Residual %	0.24
Cov Firm–Bank FE %	–0.02
<i>N</i>	48,335

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	No Controls	Relationship + Loan Controls
Std. deviation	2.87	2.87
Variance Firm FE %	0.69	0.52
Variance Bank FE %	0.17	0.16
Variance Residual %	0.24	0.22
Cov Firm–Bank FE %	–0.02	–0.03
<i>N</i>	48,335	47,891

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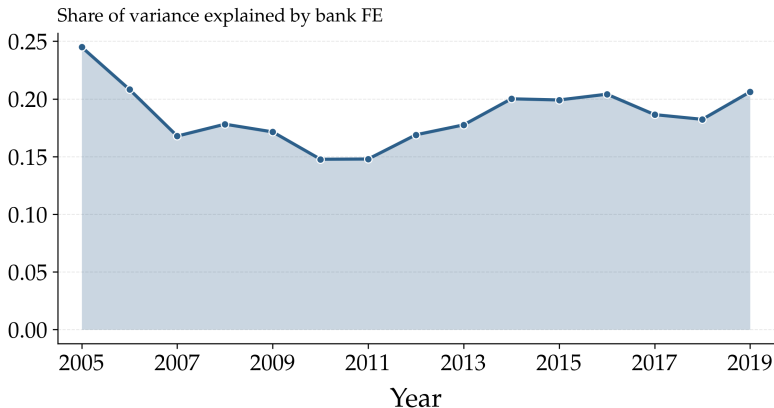
- **Additional results:**
 - sizable share of variance explained by bank FE for **other categories** of short-term loans
 - bank FE cannot be explained by **risk premia:** (National Bank FE) $\times$ (firm default risk)

Others

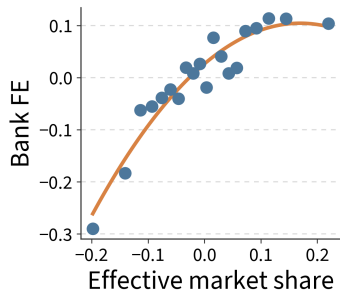
Risk

AKM Over Time

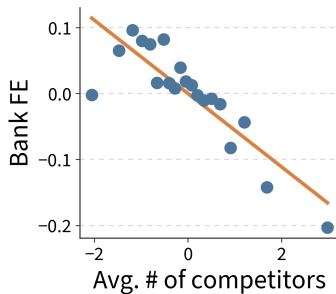
- **Bank FE over time:** computed using Italian credit register
 - estimate AKM decomposition for each year → bank-province-year FE
 - importance of bank FE ranges between **15%-25%**



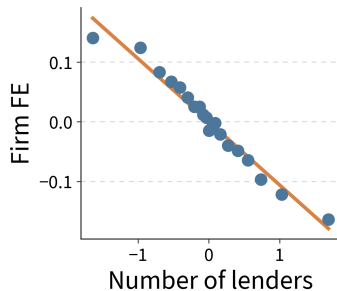
Result 2: Fixed Effects and Measures of Market Power



Bank FE $\nearrow$ **market share**



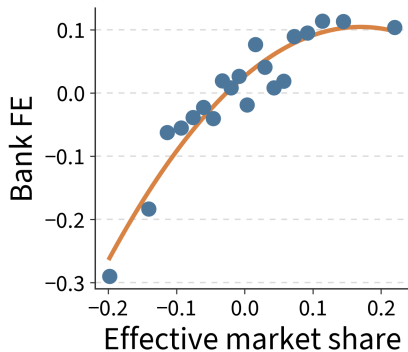
Bank FE $\searrow$ **# of competitors**



Firm FE $\searrow$ **# of lenders**

Residualized on: nat'l bank $\times$ year FE (+ province $\times$ year FE, left); firm FE, sector $\times$ year FE, firm controls (right).

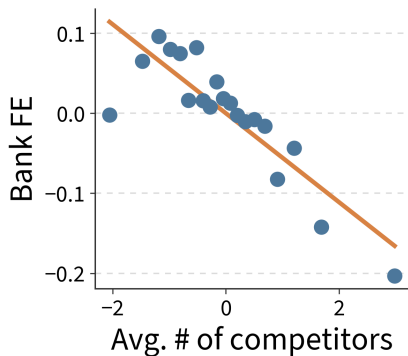
Bank FE and Market Share



	(1)	(2)	(3)	(4)	(5)
	Effective market share			Market share	
				revenue	volume
Share	1.32*** (0.07)	1.02*** (0.06)	0.87*** (0.06)	1.26*** (0.11)	0.26** (0.11)
Nat'l bank×year FE		✓	✓	✓	✓
Province×year FE			✓	✓	✓
Obs.	19,571	19,011	19,011	19,011	19,011

- Effective market share: bank's avg. share of its borrowers' total interest payments, as in Blumenfeld, Patterson and Vavra (2026)
- Market share: bank's share of total interest revenue (revenue) or total loans (volume) in the province
- Unit: bank × province × year. Figure: column (3), residualized binscatter

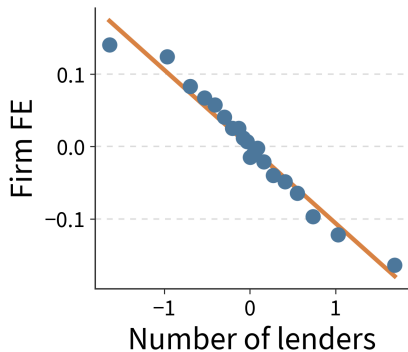
Bank FE and Number of Competitors



	(1)	(2)	(3)
# competitors	-0.041*** (0.006)	-0.040*** (0.006)	-0.050*** (0.006)
Nat'l bank FE		✓	
Year FE		✓	
Nat'l bank × year FE			✓
Bank controls		✓	
Obs.	19,571	19,288	19,011

- # competitors: number of other banks lending to the bank's borrowers (loan-weighted average)
- Unit: bank × province × year. Figure: column (3), residualized binscatter

Firm FE and Number of Lenders



	(1)	(2)	(3)
# lenders	-0.143*** (0.002)	-0.121*** (0.003)	-0.104*** (0.003)
Firm FE		✓	✓
Year FE		✓	
Sector×year FE			✓
Firm controls			✓
Obs.	724,791	669,295	649,625

- # lenders: number of banks lending to the firm
- Firm controls: log total assets, log employees, short-term debt (log and share of total assets)
- Unit: firm × year. Figure: column (3), residualized binscatter

Model

Model Overview

- **Gertler-Kiyotaki (2010) / Gertler-Karadi (2011)** + oligopolistic lending
- Economy consists of **continuum of banks** and **continuum of firms**
- Key ingredients:
 1. **Oligopolistic competition**: each firm matched with random (finite) number of banks
 2. **CES loan aggregation**: imperfect substitutability across lenders (η)
 3. **Leverage constraint**: banks face capital requirements (θ)
 4. **Bertrand-Nash**: banks set rates taking rivals' rates as given
- Dynamics: stochastic exit/entry of banks + mc shocks $\rightarrow$ stationary distribution of bank sizes
- Households: supply labor, consume and save in bank deposits, $U(C, N) = C - \chi \frac{N^{1+\psi}}{1+\psi}$

Firms

- Continuum of firms $f \in (0, 1)$ with DRTS technology
- **Working capital constraint**: fraction γ of wage bill financed by bank loans ℓ_f
- Firm f matched with **random set $B(f)$ of banks** containing a finite number N_f^b of banks
 - $N_f^b - 1 \sim \text{Bin}(\bar{B}, \pi)$
- Loans are **imperfect substitutes** (CES, elasticity η)

$$\begin{aligned} \max_{\{n_f, \ell_f, \{\ell_{fb}\}_{b \in B(f)}\}} \quad & n_f^\sigma - wn_f - \sum_{b \in B(f)} \ell_{fb} i_{fb}, \\ \text{s.t.} \quad & \gamma wn_f \leq \ell_f, \\ & \ell_f = \left(\sum_{b \in B(f)} \ell_{fb}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \end{aligned}$$

Banks

- Continuum of banks $b \in (0, 1)$; state variable: start-of-period net worth k_{bt} and servicing cost z_{bt}
- Balance sheet: loans = net worth + deposits
- Stochastic **servicing cost: z_{bt}** follows an AR(1) in logs
- Subject to **leverage constraint:** $k_b \geq \theta \int_f \ell_{fb} df$
- Random exit with probability ϕ ; replaced by new bank with k_0
- Set **firm-specific interest rates**, taking as given competitors' rates

$$\begin{aligned}
 \tilde{k}_{bt+1} = \max_{\{a_{bt}, \{i_{fbt}\}\}} & \int_{f: b \in B(f)} (i_{fbt} - z_{bt}) \overbrace{\ell(i_{fbt}, i_{f-bt})}^{\text{firm-level loan demand function}} df - i_{at} a_{bt} + k_{bt}, \\
 \text{s.t.} & \int_{f: b \in B(f)} \ell(i_{fbt}, i_{f-bt}) df \leq k_{bt} + a_{bt} \\
 & \theta (k_{bt} + a_{bt}) \leq k_{bt}
 \end{aligned}$$

Equilibrium

An equilibrium is a sequence of quantities $\{N_t, C_t, \{a_{b,t}, \ell_{fb,t}, k_{b,t+1}\}\}_t$ and prices $\{W_t, i_{a,t}, \{i_{fb,t}\}\}$ for all b, f such that given initial bank distribution on net worth and marginal cost:

1. Banks maximize profits given their financial constraint, taking as given competitors' interest rates and aggregate quantities W, C
2. Firms maximize profits
3. Households maximize utility, subject to budget constraint
4. All markets clear

Optimal Pricing

- Loan demand: $\ell_{fb} = \left(\frac{i_{fb}}{i_f}\right)^{-\eta} \ell_f$, $\ell_f = \gamma w \left(\frac{\sigma}{w(1+\gamma i_f)}\right)^{\frac{1}{1-\sigma}}$, $i_f = \left(\sum_{b \in B(f)} i_{fb}^{1-\eta}\right)^{\frac{1}{1-\eta}}$

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Bank's optimal rate for firm f

$$i_{fb} = \frac{\varepsilon(i_{fb}; i_f)}{\varepsilon(i_{fb}; i_f) - 1} \cdot mc_b$$

- mc_b : shadow marginal cost of funds ($mc_b > i_a + z_b$ if leverage constraint binds)

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- mc_b : shadow marginal cost of funds ($mc_b > i_a + z_b$ if leverage constraint binds)
- $\varepsilon(i_{fb}; i_f)$: **loan demand elasticity** driven by two forces:

$$\varepsilon(i_{fb}; i_f) = \underbrace{\eta(1 - s_{fb})}_{\text{credit substitution: } d(\ell_{fb}/\ell_f)} + \underbrace{\frac{1}{1-\sigma} \cdot \frac{\gamma i_f}{1+\gamma i_f} \cdot s_{fb}}_{\text{credit demand: } d\ell_f}, \quad s_{fb} = \frac{\ell_{fb} i_{fb}}{\sum_{b' \in B(f)} \ell_{fb'} i_{fb'}}$$

AKM Decomposition in the Model

- Model can generate bank and firm FE in AKM-style regression
 - **firm FE**: random matching, some firms have access to more banks
 - **bank FE**: banks have different mc_b and charge different markups

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- Log pricing: $\ln i_{fb} = \underbrace{\ln \mu_{fb}}_{\text{markup}} + \underbrace{\ln mc_b}_{\text{marginal cost}}$

- Model generates both firm FE and bank FE:

$$\ln i_{fb} = \bar{\tau} + \underbrace{(\mathbb{E}_f[\ln \mu_{fb}] - \mathbb{E}[\ln \mu_{fb}])}_{\text{Firm FE: avg markup faced by firm}} + \underbrace{(\ln mc_b - \mathbb{E}[\ln mc_b]) + (\mathbb{E}_b[\ln \mu_{fb}] - \mathbb{E}[\ln \mu_{fb}])}_{\text{Bank FE: cost heterogeneity + markup heterogeneity}}$$

Quantitative Analysis

Solving the Model

- **Challenge:** firm-level Bertrand-Nash interest rates have **no closed form** solution
 - Each firm is a separate pricing game among its N_f^b lenders
 - Each bank plays a continuum of such games, linked by its balance sheet
- **Key idea:** given mc_b , bank's problem separates across its borrowers
 - Interest rates faced by firm f depend only on its lenders' costs $\{mc_b\}_{b \in B(f)}$
 - **Solve all pricing games once**, independent of net-worth distribution and the wage

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 - **Solve all pricing games once**, independent of net-worth distribution and the wage
- **Algorithm:**
 1. **Solve $i_{fb}(mc_b, mc_{-b}) = \frac{\varepsilon_{fb}}{\varepsilon_{fb}-1} mc_b$ on a grid of lenders' cost vectors**
 2. Find the stationary distribution of banks $\Omega(k, z)$ and the mapping to marginal cost $mc(k, z)$
 - Guess $\Omega(k, z)$ and $mc(k, z)$
 - Use $i_{fb}(\cdot)$ to compute expected loans $L(mc)$ integrating over competitors
 - Update $mc(k, z)$ from leverage constraint: $k = \theta L(mc)$ if binding, else $mc = i_a + z$
 - Update $\Omega(k, z)$ with net worth law of motion, exit and entry; repeat
 3. Recover the wage in closed form from labor market clearing

Calibration

- **Target:** loan demand elasticities, concentration, N. bank relationships

Param.	Description	Value	Target	Data	Model
<i>Estimated (SMM)</i>					
η	Loan CES index	4.4	Total loan semi-elasticity →	−19	−19
σ	Returns to scale	0.98	Substitution semi-elasticity →	−12	−12
ϕ	Exit probability	0.12	Av. s_{fb} of 1st largest bank (%)	83	83
k_0	Initial net worth	10^{-6}	Av. s_{fb} of 2nd largest bank (%)	19	19
π	Matching prob.	0.43	Avg. banks/firm	2.3	2.3
ρ_z	Persistence of z_{bt}	0.51	Persistence of s_b	0.73	0.89
σ_z	Std. deviation z_{bt} shocks	0.81	Std. deviation of Δs_b	0.10	0.11
<i>Calibrated externally</i>					
θ	Min. net-worth ratio	0.08	Basel capital requirements		
β	Discount factor	0.99	Standard		
ψ	Frisch elasticity	1	Standard		
χ	Labor disutility	1	Normalize output		
γ	Working capital ratio	0.45	Short-term debt/revenue		
$\bar{B} + 1$	Max. number of banks per firm	4	90th percentile		

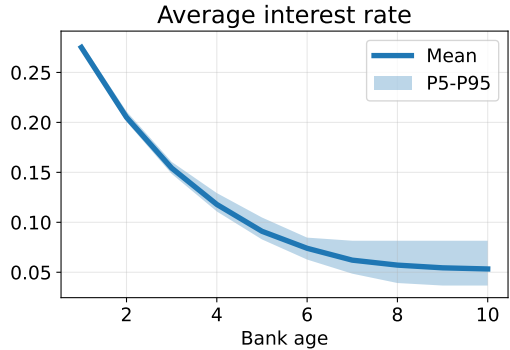
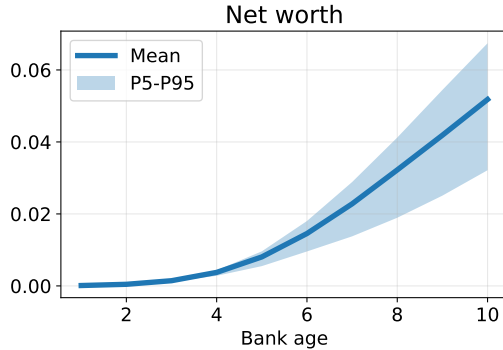
Loan Demand Elasticities

- From the model we have **structural elasticities**

$$\left. \frac{\partial \log(\ell_{fb})}{\partial \log(i_{fb})} \right|_{i_{f,-b}} = -\eta [1 - s_{fb}] - \frac{1}{1 - \sigma} \left[\frac{\gamma i_f}{1 + \gamma i_f} \right] s_{fb}$$
$$\left. \frac{\partial \log(\ell_{fb}/\ell_f)}{\partial \log(i_{fb})} \right|_{i_{f,-b}} = -\eta [1 - s_{fb}]$$

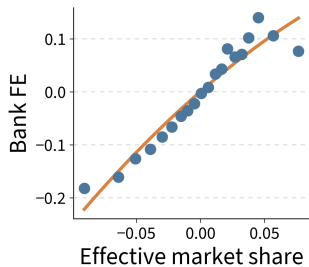
- Strategic interactions:** structural elasticities $\neq$ reduced-form elasticities ($i_{f,-b}$ not constant)
- Use credit register to estimate **reduced-form elasticities**
 - instrument Δi_{fb} with the change in the average rate charged by bank b
Amiti and Weinstein (2018)
- Indirect inference:** estimate reduced-form elasticities in the model 

Bank Life-Cycle Dynamics

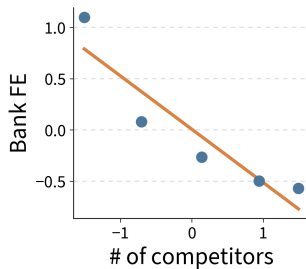


Stationary distribution

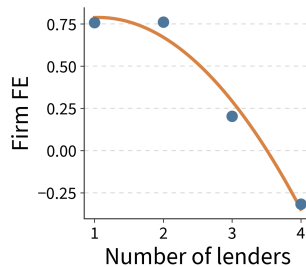
Model Validation



Bank FE $\nearrow$ market share



Bank FE $\searrow$ n. competitors



Firm FE $\searrow$ # of lenders

Counterfactual: Cost of Market Power

- **Competitive pricing:** set $i_{fb} = i_a + z_b$ for all bank-firm pairs
 - lower rates i_{fb} reduce firms' marginal cost and increase output

Output effects of eliminating market power

- **Short run (year 1): +1.2%**
- **Long run : -6.4%**

markup ↓
mc ↑

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- Why does the effect **flip over time**?
 1. Eliminate markups → **no bank profits** → erodes net worth
 2. Tighter leverage constraints → higher mc_b → **overturns short-run rate decline**

Macroprudential Policy and Bank Competition

- **Reduce capital requirement:** $\theta = 8\% \rightarrow 4\%$

	1 year	20 years
Output (%)	+0.40	+0.69
of which: less misallocation	+0.10	+0.11
Average rate (pp)	−0.45	−0.60
of which: cost of funds	−0.24	−0.32
of which: markups	−0.21	−0.28
Market share of top 30% of banks (pp)	−12	−14

- **Result:** ~**half** of rate decline from **lower markups** (not lower funding costs)
- Small banks expand $\rightarrow$ less concentrated loan market $\rightarrow$ lower markups

Conclusion

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- **Bank market power in corporate loan markets**

- ▶ Bank FE: 15–25% of interest rate variation
- ▶ Strongly correlated with local market shares

- **GE model with oligopolistic banks**

- ▶ Fits loan demand elasticities, concentration, bank relationships
- ▶ Reproduces untargeted FE–market power relationships

- **Counterfactuals**

- ▶ Eliminating market power: **+1.2% (SR), —6.4% (LR)**
- ▶ Capital requirements: **half of interest rate response from endogenous markups**

Thank You!

AKM Results: Other Loan Categories

- **Robustness:** similar results for other categories of short-term debt

AKM

	Variance share: bank FE	Outstanding amount (% of total)	Share with a loan
Liquidity credit	0.16	13	58
Irrevocable credit lines	0.18	7	24
Receivable-based	0.17	41	59
Term loans	0.12	27	27
Trade financing	0.11	12	10

Alternative Interpretation: Risk Premia

Concern: the bank-province FE may capture **risk premia**, not market power.

AKM

- Baseline residualization controls for default risk:

$$r_{lib,t} = \delta'_{pd} pd_{ib,t} + \delta'_{col} collateral_{lib,t} + \tilde{r}_{lib,t}.$$

- But different banks may price the same default risk differently.
 - Then the bank-province component could partly reflect bank-specific risk premia.
- **Robustness checks:**
 - Absorb **national-bank** $\times$ **credit-risk** interactions in the first stage.
 - Estimate AKM decompositions separately for high- and low-risk firms.
- **Result:** the bank-province component remains sizable.
 - Bank-province FE explains 10% of variance *after* absorbing national bank FE and risk premia
 - Similar patterns for high- and low-risk firms.

Data coverage: AnaCredit and TAXIA

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- **AnaCredit**

- Harmonized euro-area credit register
- All loans above €25,000 extended by credit institutions in the euro area
- Rate: contractually agreed interest rate at the loan-level
- Time: Available from 2018 onward
- Banks: with national market share $< 2\%$ are exempted from reporting
- Banks: with national market share $< 4\%$ can be temporarily exempted from reporting

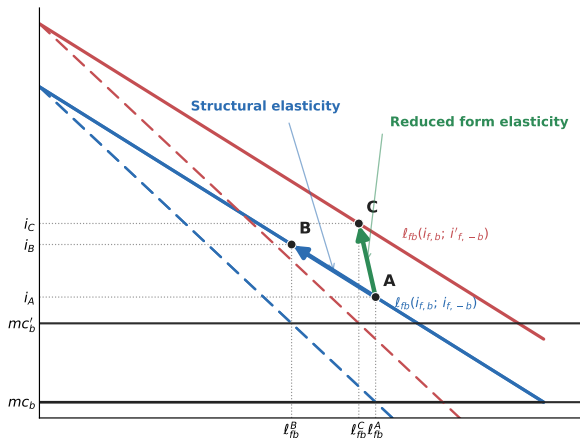
- **ITAXIA**

- Italian Credit Register
- Firm-bank-quarter information on credit above €75,000
- Rate: calculated from reported interest payments
- Time: longer time series (2005–2019), including small banks
- Banks: better coverage of small local banks

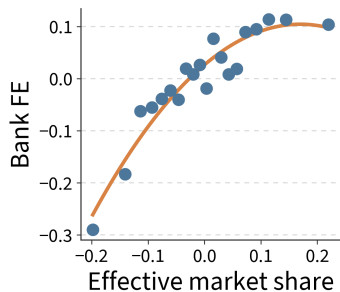
Structural vs Reduced-form Elasticities

- **Structural elasticities:** keep constant $i_{f,-b}$
- **Reduced-form elasticities:** other banks best-respond and $i_{f,-b} \rightarrow i'_{f,-b}$

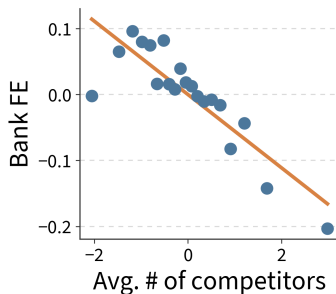
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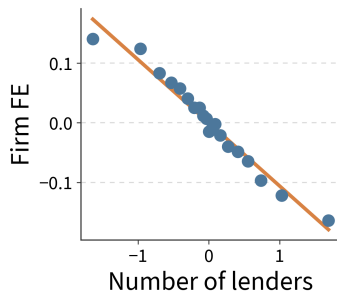
Result 2: Fixed Effects and Measures of Market Power



Bank FE  **market share**



Bank FE  **# of competitors**



Firm FE  **# of lenders**

Residualized on: nat'l bank $\times$ year FE (+ province $\times$ year FE, left); firm FE, sector $\times$ year FE, firm controls (right).

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Stationary Distribution of Banks

